

F O R T I M A N A G E R N S E 6

Lab 05 — Device Database, Installation and Synchronization

01. The Device Database

Lab 04 established a managed relationship. This lab is about what that relationship actually does: FortiManager keeps its own copy of the FortiGate configuration, and every change — in either direction — is a transfer between that copy and the live device.

The whole of FortiManager rests on this one idea. Policy packages, provisioning templates, revision history and the install preview are all operations against the stored copy rather than against the firewall, and almost every confusing behavior in the product comes from forgetting which of the two is being looked at.

The lab continues on the pair that registered successfully in Lab 04 — FortiManager 7.4.3 and FortiGate 7.2.1 (fortigate7-2, 10.58.228.177) in the root ADOM, starting from connectivity up and configuration status Synchronized.

► 1.1 What the interface is showing

Opening a managed device in FortiManager presents what looks like the FortiGate GUI — System Settings, Network, VPN, Log & Report. It is not the FortiGate. It is FortiManager's stored copy of that device's configuration, known as the device database.

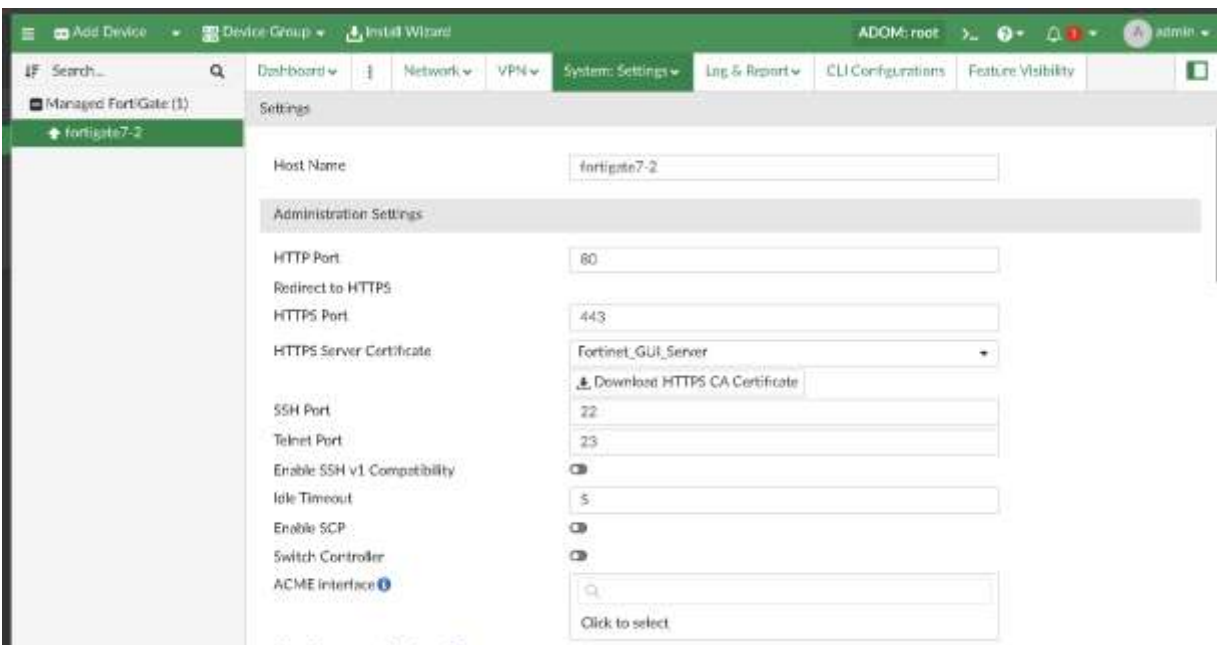


Figure 1 — The device configuration as presented inside FortiManager — a copy, not the live device.

Two consequences follow, and both matter for the rest of the series.

- **Editing here** changes nothing on the firewall. The change sits in the database until it is installed.
- **A change made on the firewall** is not here either, until FortiManager retrieves it.

Those two directions — install and retrieve — are the whole of this lab. Sections 02 and 03 take one each.

Why it is built this way

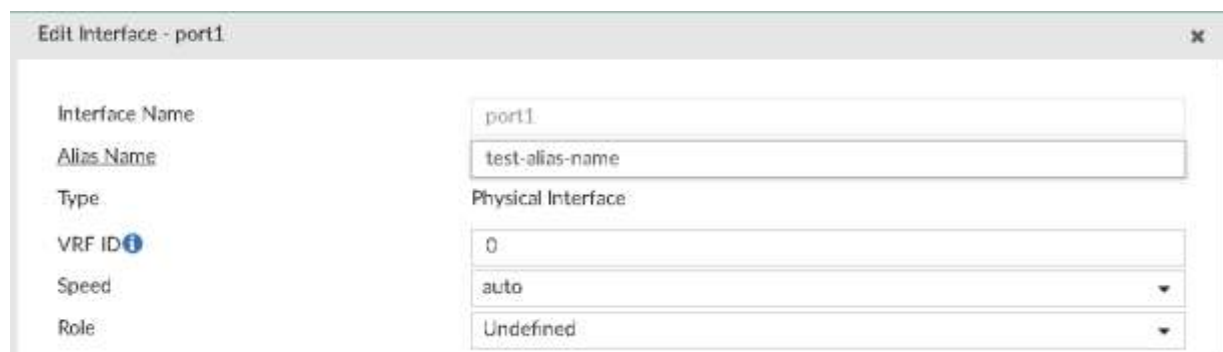
Keeping a separate copy is what makes staged change possible: an administrator can prepare a configuration, have it reviewed, preview exactly what will be sent, and install it at a chosen time.

It is also what makes the out-of-sync states possible, because two copies can always disagree.

02. Changes Made in FortiManager

▸ 2.1 Making the change

An interface alias was set on port1 from inside the device database. The alias is a deliberately harmless change — visible in a diff, and carrying no risk of losing management access if the install behaves unexpectedly.



Interface Name	port1
Alias Name	test-alias-name
Type	Physical Interface
VRF ID ⓘ	0
Speed	auto
Role	Undefined

Figure 2 — Setting an alias on port1 inside the device database.

▸ 2.2 The device goes Modified

The device list immediately reports configuration status Modified. The firewall has not been contacted; this status describes the database, and specifically that it now holds something the device does not.



☐	Device Name ⇅	Config Status ⇅	Host Name ⇅	IP Address ⇅	Platform ⇅	Description ⇅	
☐	fortigate7-2	Modified	fortigate7-2	10.58.228.177	FortiGate-VM64-...		

Figure 3 — Configuration status showing Modified after the database edit.

▸ 2.3 The Install menu

Three entries sit under Install, and the difference between them is a frequent exam point.



☐	Device Name ⇅	Config Status ⇅	IP Address ⇅	Platform ⇅	Description ⇅	
☒	fortigate7-2	Modified	10.58.228.177	FortiGate-VM64-...		

Figure 4 — Install Wizard, Quick Install (Device DB) and Re-install Policy.

Entry	What it sends	When to use it
Install Wizard	Policy package and/or device settings, with a preview step and device selection	The normal path — the only one that shows what will be sent before sending it
Quick Install (Device DB)	Device-level settings only, with no preview and no policy package	Small device-setting changes where the content is already known
Re-install Policy	The policy package as last installed	Pushing an unchanged package back to a device. Greyed out here, because no package has ever been installed

▸ 2.4 Choosing what to install

The wizard separates the policy package from device settings. Only device settings are installed here, since the change was an interface alias and no policy package exists yet in this ADOM.

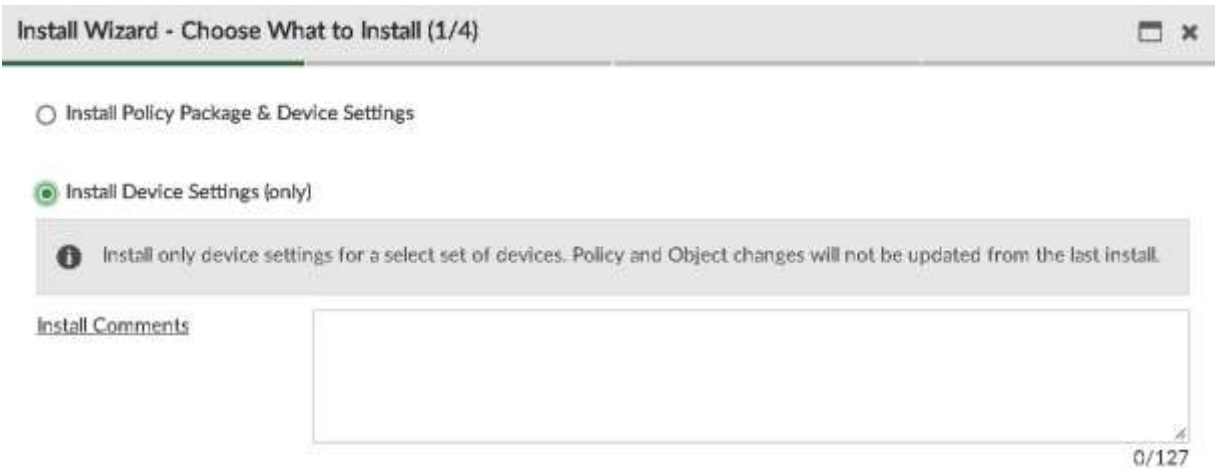


Figure 5 — Install Wizard step 1 — Install Device Settings (only) selected.

The install comment field is worth using in practice even though it was left empty here: it is recorded against the resulting revision and is the cheapest form of change documentation available in the product.

▸ 2.5 Selecting the device



Figure 6 — Install Wizard step 2 — device selection.

▸ 2.6 The install preview

Before anything is sent, the wizard renders the change as the exact CLI that will be executed on the device — here, three lines setting the alias.



Figure 7 — Install preview rendering the pending change as FortiGate CLI.

The preview is a diff, not a summary

It is the difference between the device database and the device, expressed as commands. If it shows more than expected, something else was modified in the database earlier and is about to be installed alongside the intended change.

Reading it is the habit that prevents most accidental installs. Quick Install skips this screen entirely.

2.7 Installing

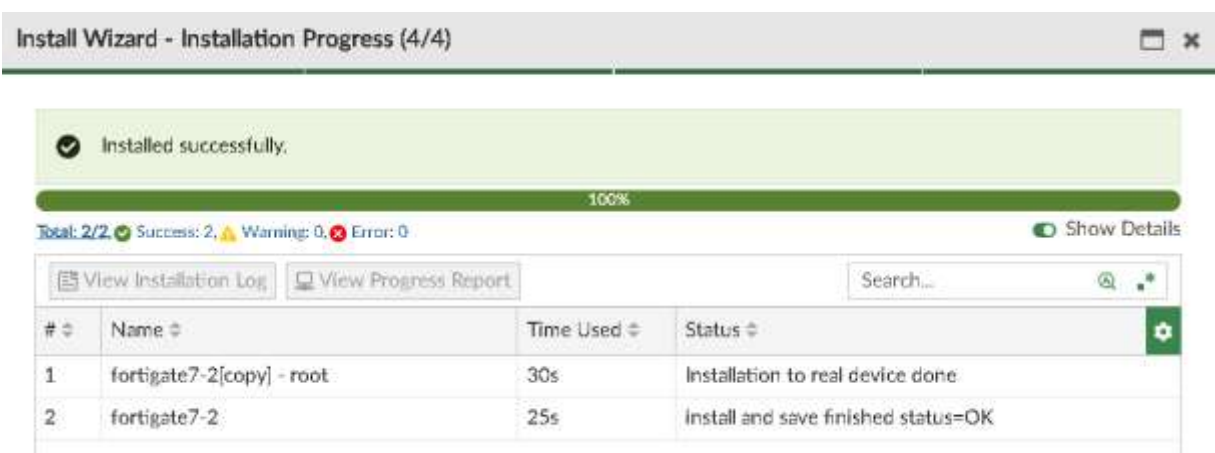
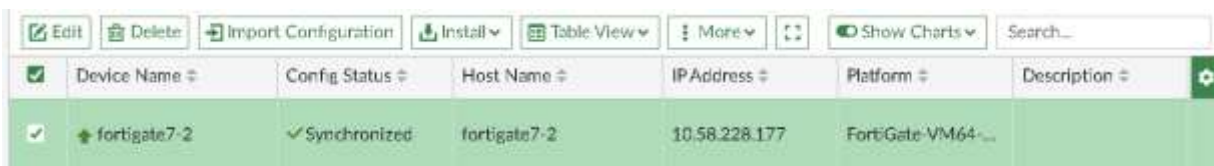


Figure 8 — Installation completed — two tasks, both successful.

Observation	A single install produced two tasks: installation to the real device, then an install-and-save status returned by the device.
Why	The first is FortiManager sending the configuration; the second is the FortiGate confirming it applied and saved it.
Risk	A device can show the first succeeding and still not be in sync. Both have to succeed before the device is considered synchronized.

▸ 2.8 Back to Synchronized



☒	Device Name	Config Status	Host Name	IP Address	Platform	Description
☒	fortigate7-2	✓ Synchronized	fortigate7-2	10.58.228.177	FortiGate-VM64...	

Figure 9 — Configuration status returned to Synchronized.

Synchronized means the database and the device agreed as of the last install or retrieve. It is not a continuous live comparison.

03. Changes Made on the FortiGate

The opposite direction was then tested by changing something larger directly on the firewall: its hostname.

▸ 3.1 The managed-device warning

The FortiGate warns before accepting the change, and the wording repays reading closely.



Figure 10 — FortiGate warning that a local change will put the device out of sync.

The most consequential behaviour in this lab

A local change to a system setting survives. A local change to a firewall policy or object is temporary and will be silently overwritten the next time a policy package is installed.

This is the reason emergency changes made on a managed firewall during an incident have to be re-made in FortiManager afterwards — and the reason undocumented local change is a quiet source of outage.

3.2 Changing the hostname

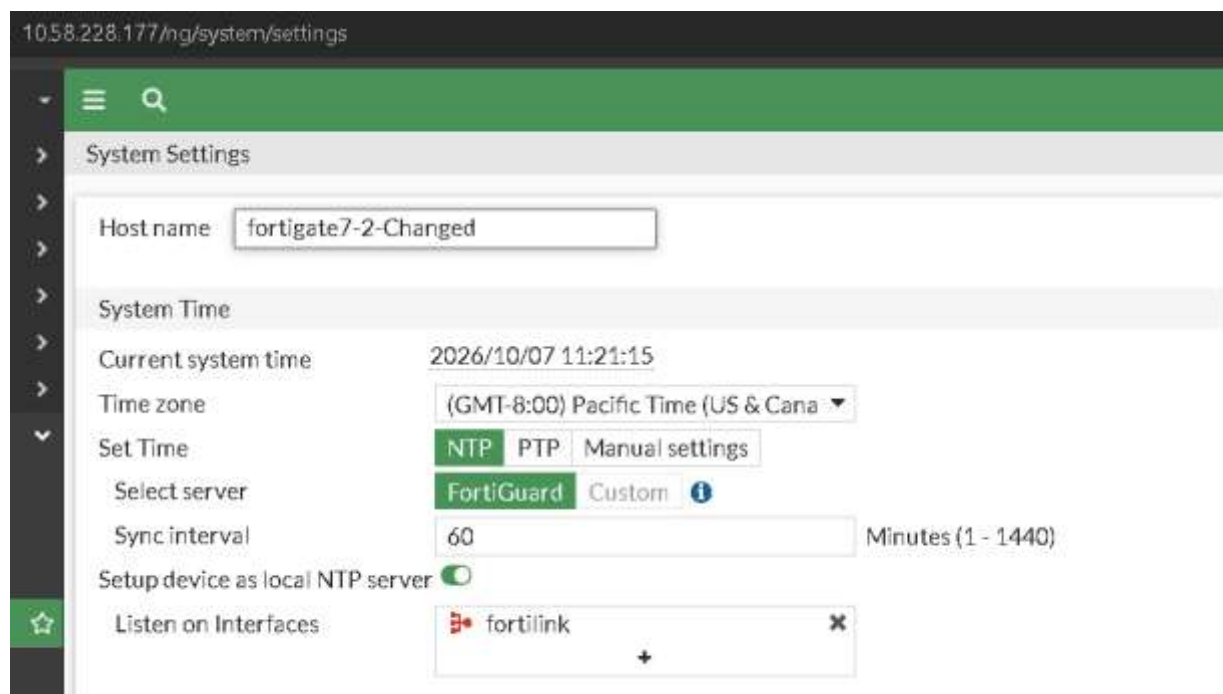


Figure 11 — Hostname changed to fortigate7-2-Changed on the FortiGate.

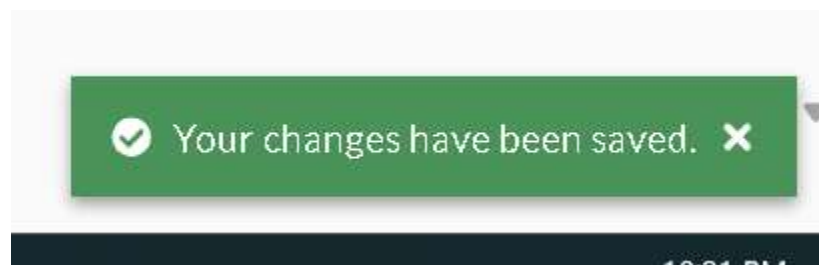
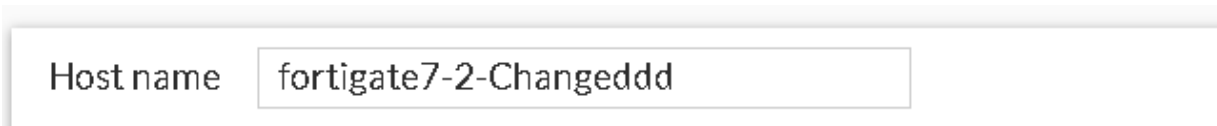


Figure 12 — Change saved on the device.

A second change was then made, to confirm the behaviour repeats rather than being a one-off retrieval.



Host name

Figure 13 — Hostname changed again on the FortiGate.

▸ 3.3 Automatic retrieval

FortiManager picks the change up without being asked. The device list shows the new hostname and a configuration status of Modified (recent auto-updated).



☐	Device Name ▾	Config Status ▾	Host Name ▾	IP Address ▾	Platform ▾	De 
☐	 fortigate7-2	 Modified (recent auto-updated)	fortigate7-2-Changedddd	10.58.228.177	FortiGate-VM64-...	

Figure 14 — Status showing Modified (recent auto-updated) with the new hostname.

The parenthetical is the important part. Plain Modified means the database holds a change waiting to go to the device. **Modified (recent auto-updated) means the opposite — the database was updated from the device, and the change originated there.** The same word covers both directions, which is why the status alone is not enough to tell what happened; the qualifier is.

04. Task Monitor

Every operation in this lab is recorded under System Settings > Task Monitor with its source, description, initiating user, result, duration and ADOM.

ID	Source	Description	User	Status	Time Used	ADOM	Start Time	End Time
15	Device Manager	Update Device	admin	Success: 1	3s	root	Wed, Oct 7, 2026 11:22 A...	Wed, Oct 7, 2026 11:22 A...
14	Install Preview	Install Preview	admin	Success: 1	1s	root	Wed, Oct 7, 2026 11:21 A...	Wed, Oct 7, 2026 11:21 A...
13	Install Device	Install Device	admin	Success: 2	3s	root	Wed, Oct 7, 2026 11:21 A...	Wed, Oct 7, 2026 11:21 A...
12	Install Device	Install Device	admin	Success: 2	32s	root	Wed, Oct 7, 2026 11:18 A...	Wed, Oct 7, 2026 11:18 A...
11	Install Preview	Install Preview	admin	Success: 1	9s	root	Wed, Oct 7, 2026 11:17 A...	Wed, Oct 7, 2026 11:17 A...
10	Install Device	Install Device	admin	Success: 2	2s	root	Wed, Oct 7, 2026 11:17 A...	Wed, Oct 7, 2026 11:17 A...
9	Install Device	Install Device	admin	Success: 2	13s	root	Wed, Oct 7, 2026 11:11 A...	Wed, Oct 7, 2026 11:11 A...
8	Import Wizard	Import Device Obj/Policy	admin	Success: 1	2s	root	Wed, Oct 7, 2026 11:05 A...	Wed, Oct 7, 2026 11:05 A...
7	Device Manager	Add Device	admin	Success: 1	19s	root	Wed, Oct 7, 2026 10:21 A...	Wed, Oct 7, 2026 10:21 A...
6	Device Manager	Add Multiple Devices	admin	Error: 3	4m 51s	7_2ADOM	Wed, Oct 7, 2026 10:16 A...	Wed, Oct 7, 2026 10:16 A...
5	Device Manager	assign adom 7_2ADOM object member	admin	Success: 1	2s	7_2ADOM	Wed, Oct 7, 2026 10:14 A...	Wed, Oct 7, 2026 10:14 A...
4	Firmware Manager	Download Object	admin	Success: 1	1m 30s	root	Wed, Oct 7, 2026 9:36 AM...	Wed, Oct 7, 2026 9:36 AM...

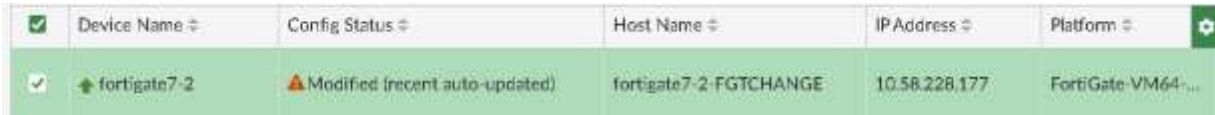
Figure 15 — Task Monitor — the full operation history including the installs from section 02.

Reading the capture against the work done: the Install Preview and Install Device entries are the wizard runs from section 02, Update Device is the automatic retrieval from section 03, and Add Device is the Lab 04 registration.

Observation	Task 6, Add Multiple Devices against 7_2ADOM, is logged as an error — and task 7, Add Device against root, succeeded immediately afterwards.
Why it matters	This resolves the open question from Lab 04, where the device had been assigned to 7_2ADOM during authorization but reported adom:root afterwards. The ADOM assignment did not silently change; it failed, and the Discover-wizard addition placed the device in root.
Habit	Task Monitor is the first place to look when an operation appears to have done nothing. A failed task is recorded with the same prominence as a successful one, and nothing else in the interface reports it.

05. Competing Changes and Change Acceptance

The last test was to change the hostname on both sides and see which copy prevails.



☒	Device Name ▾	Config Status ▾	Host Name ▾	IP Address ▾	Platform ▾	
☒	fortigate7-2	Modified (recent auto-updated)	fortigate7-2-FGTCHANGE	10.58.228.177	FortiGate-VM64...	

Figure 16 — Status after a further change made on the FortiGate.

What the capture establishes is that the FortiGate change was retrieved and the database now carries the device's hostname, with the status again showing Modified (recent auto-updated). Which copy would survive an install, and what happens when both sides hold genuinely different pending values, is not established by this capture alone.